

Propagation modelling of dispersion and discussion how this relates to the 2018 Nobel Prize for Chirped Pulse Amplification

Ali Fadlallah

Master CompuPhys, EIPHI Graduate School, Université de Bourgogne-Franche-Comté
`ali.fadlallah@edu.univ-fcomte.fr`

October 15, 2023

Contents

1	Introduction	3
2	Overview of CPA	3
3	Numerical Simulation for Dispersion in CPA	4
4	Results	6
4.1	Power	6
4.2	Phase	7
4.3	Chirp	8
4.4	Dispersion Only Output	8
4.5	Dispersion and Nonlinearity	9
4.5.1	Power	9
4.5.2	Chirp	10
5	Conclusion	11

1 Introduction

The generation of high peak power laser pulses has been a significant goal in laser physics since the invention of the Laser in 1960. Traditional techniques faced limitations in reaching intensities beyond the gigawatt threshold due to nonlinear effects and potential damage to the gain medium [1]. To overcome these challenges, researchers turned to radar technology, which employed dispersion for pulse stretching and compression.

In the mid-1980s, Donna Strickland and Gerard Mourou introduced Chirped Pulse Amplification (CPA), a groundbreaking technique inspired by radar principles, into the optical regime. CPA has the potential to generate ultrashort pulses with energies at the Joule level. By breaking the gigawatt intensity limit, Strickland and Mourou's work opened new avenues for laser physics [2].

In recognition of their groundbreaking contribution, Donna Strickland and Gerard Mourou were awarded the Nobel Prize in Physics in 2018. Their achievement in developing Chirped Pulse Amplification revolutionized laser technology, enabling the generation of ultrahigh peak power pulses.

This project focuses on the propagation modeling of dispersion and its relation to the 2018 Nobel Prize for Chirped Pulse Amplification. Through numerical simulations and analysis, we aim to study the behavior of laser pulses in the presence of dispersion and evaluate the effectiveness of CPA in mitigating its effects. The findings will contribute to a deeper understanding of dispersion and its implications for ultrashort pulse generation.

2 Overview of CPA

The self-focusing phenomenon posed a challenge to amplifying ultra-short laser pulses. The solution was introduced through Chirped Pulse Amplification (CPA). In this technique:

- An initial laser pulse with high peak power and short duration is generated using a mode-locked laser oscillator.
- To overcome self-focusing, the pulse is stretched using a dispersive medium, such as diffraction gratings.
- Diffraction gratings offer several advantages in pulse stretching:
 - Broad spectral bandwidth
 - Precise control over dispersion
 - High diffraction efficiency
 - Compact and versatile design
 - Compatibility with a wide range of wavelengths
- The pulse stretching introduces controlled group velocity dispersion (GVD), causing different frequency components within the pulse to spread out in time, resulting in a longer-duration pulse known as a chirped pulse.
- The stretched pulse is then amplified to access higher energies without encountering self-focusing.
- The amplifying medium in CPA must satisfy three conditions:
 - Sufficiently broad bandwidth to encompass the entire spectrum of the pulse, preventing distortion.
 - Fluence of the pulse approaching the saturation fluence of the amplifying medium, maximizing energy utilization.
 - Intensity within the amplifier staying below a critical level to avoid significant nonlinear effects that can distort the pulse's spatial and temporal profiles.
- To restore the pulse to its original duration and compensate for the introduced positive chirp during CPA, a similar grating pair can be used.

- The grating pair possesses complementary dispersion properties, allowing for the reversal of temporal spreading and compression of the pulse.
- This compression enables the generation of high-energy, ultrashort laser pulses with high peak power.

Figure 1 illustrates the operation of Chirped Pulse Amplification.

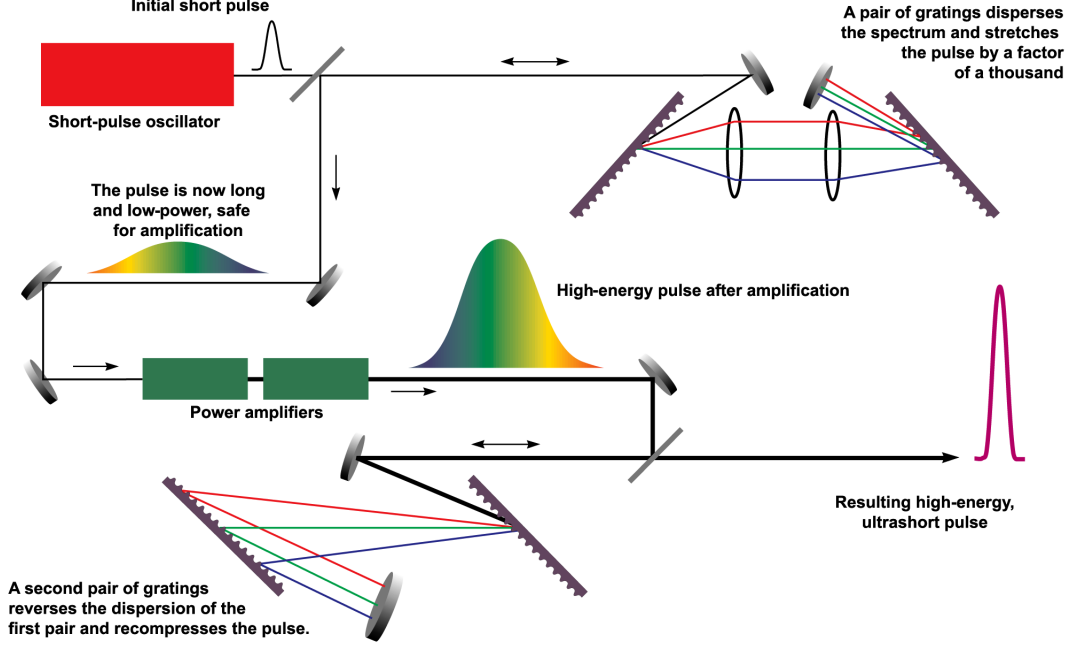


Figure 1: Chirped pulse amplification operation [3]

Certainly! Here's the section with the equations included:

3 Numerical Simulation for Dispersion in CPA

The dispersion is a crucial step in the Chirped Pulse Amplification process to stretch the input pulse and mitigate the involvement of nonlinear effects that can damage the medium and distort the output pulse. The propagation of dispersion is computed numerically using the non-linear Schrödinger's equation:

$$i \frac{\partial A(z, T)}{\partial z} = \frac{\beta_2}{2} \frac{\partial^2 A(z, T)}{\partial T^2} - \gamma |A(z, T)|^2 A(z, T) \quad (1)$$

Here, $A(z, T)$ represents the amplitude of the wave's envelope, z is the propagation coordinate, β_2 represents group velocity dispersion, and γ is the nonlinearity parameter. The first term on the right-hand side describes dispersion, while the second term accounts for nonlinearity in the temporal domain.

The short pulses associated with broad frequency spectra exhibit variation in the propagation constant (dependent on refractive index) across the pulse's spectral bandwidth:

$$\beta(\omega) = n(\omega) \frac{\omega}{c} = \beta_0 + \beta_1(\omega - \omega_0) + \frac{1}{2} \beta_2(\omega - \omega_0)^2 + \dots \quad (2)$$

In particular, we are interested in:

- $\beta_1 = \frac{\partial \beta}{\partial \omega}$, representing the reciprocal of the group velocity ($\frac{1}{v_g}$) measured in ps/km.
- $\beta_2 = \frac{\partial^2 \beta}{\partial \omega^2}$, representing group velocity dispersion (measured in ps/km·THz or ps²/km).
- ω_0 , the central frequency.

Dispersion arises from the frequency-dependent refractive index, causing different frequency components to propagate at different speeds. For the purpose of calculating dispersive propagation, we neglect the nonlinearity term and focus solely on dispersion:

$$i \frac{\partial A(z, T)}{\partial z} = \frac{\beta_2}{2} \frac{\partial^2 A(z, T)}{\partial T^2} \quad (3)$$

An analytical solution exists for this differential equation in the frequency domain:

$$\tilde{A}(z, \omega) = \tilde{A}(0, \omega) \exp\left(\frac{i}{2} \beta_2 \omega^2 z\right) \quad (4)$$

This solution of the non-linear Schrödinger's equation leads to pulse broadening and the development of quadratic temporal phase, or linear "chirp."

For an incident Gaussian pulse:

$$A(0, T) = \exp\left(-\frac{T^2}{T_0^2}\right) \quad (5)$$

The dispersive propagation in the temporal domain can be described as:

$$A(z, T) = A_0(z) \exp\left(-\frac{T^2}{2T(z)^2}\right) \exp(i\phi(z, T)) \quad (6)$$

where:

$$A_0 = \frac{1}{\sqrt{1 + \left(\frac{z}{L_D}\right)^2}} \quad (7)$$

represents the peak amplitude as a function of z . It accounts for the change in peak amplitude due to dispersion effects. Here, $L_D = \frac{T_0^2}{|\beta_2|}$ is the dispersive length, denoting the distance over which dispersion effects become significant in a medium. It represents the distance at which the pulse experiences noticeable broadening or spreading in time due to dispersion.

The size of the pulse as a function of z is given by:

$$T(z) = T_0 \sqrt{1 + \left(\frac{z}{L_D}\right)^2} \quad (8)$$

It accounts for the temporal broadening of the pulse due to dispersion.

The phase term $\phi(z, T)$ in Equation 6 represents the changes in the pulse's phase as it propagates through a dispersive medium. It influences the temporal and spectral characteristics of the pulse, as well as the overall waveform shape. It is defined as:

$$\phi(z, T) = -\frac{\text{sgn}(\beta_2)(z/L_D)}{1 + \left(\frac{z}{L_D}\right)^2} \frac{T^2}{2T(z)^2} + \frac{1}{2} \tan^{-1}\left(\text{sgn}(\beta_2) \frac{z}{L_D}\right) \quad (9)$$

The phase term considers the sign of the dispersion parameter β_2 to account for anomalous (positive) or normal (negative) dispersion.

Figure 2 illustrates the quadratic spectral phase modulation and the modification in the temporal intensity resulting from dispersion.

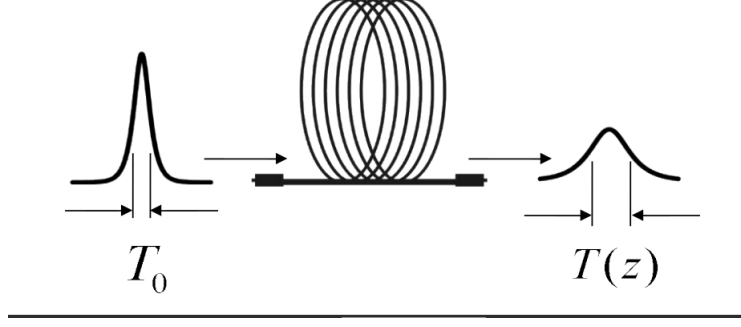


Figure 2: Quadratic spectral phase modulation and modification in the temporal intensity

4 Results

In this section, the dispersion propagation of the initial Gaussian pulse described by equation (5) will be presented through various figures produced by the provided MATLAB code.

4.1 Power

Figure 3 depict the stretching of the pulses as they propagate through the stretcher. The evolution of the pulse is governed by equation (6). The input pulse $A(0, T)$ is shown at $z = 0$, where it has the maximum amplitude and the shortest time broadening. As the pulse propagates through the dispersive stretcher, a decrease in the maximum amplitude $A_0(z)$ and an increase in the duration $T(z)$ (pulse size) are observed, leading to the stretching of the input pulse.

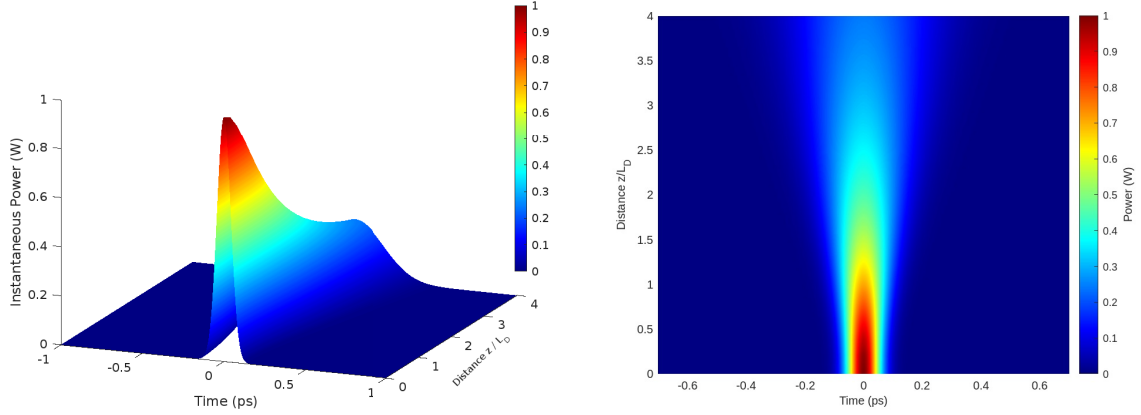


Figure 3: Propagation Modeling and Dispersion Analysis of Gaussian Pulses with Distance. Left plot: 3D representation of pulse dispersion with z . Right plot: 2D representation.

Figure 4 provide information about how the maximum amplitude decreases with distance and how the pulse size changes during propagation. In the first figure, the behavior of the maximum amplitude of the pulses follows equation (7). At the normalized distance $z/L_D = 4$, the maximum amplitude decreases to 0.242 of its initial value. In the second figure, the pulse size increases according to equation (8). At the normalized distance $z/L_D = 4$, the full width at half maximum (FWHM) of the pulse increases to 4.12 times its initial value.

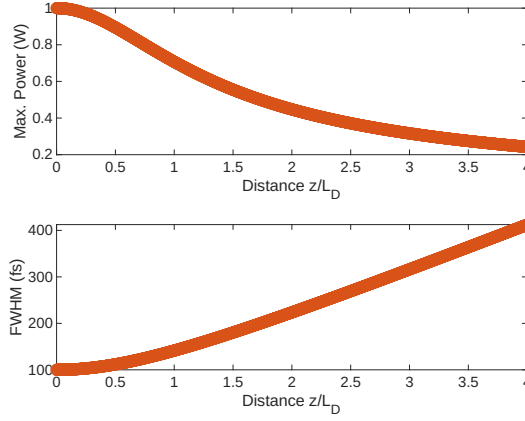


Figure 4: Variation of Peak Power and FWHM with Distance in Pulse Propagation

4.2 Phase

The phase variation in pulse propagation can be understood by examining equation (9). Initially, as the pulse enters the medium at $z = 0$, the phase change is positive, $\phi(z, T) > 0$ (for $0 < z < 0.5$). This positive phase change is represented by the term $\frac{1}{2} \tan^{-1} \left(\text{sgn}(\beta_2) \frac{z}{L_D} \right)$, contributing to the initial increase in phase and establishing the reference phase for subsequent changes.

However, as the pulse propagates further, the dominant term in the phase equation becomes proportional to z/L_D . This term leads to a change in the phase sign, transitioning from positive to negative values (as shown in Figure 5). The sign change occurs when the distance traveled by the pulse z becomes comparable to or exceeds the characteristic dispersion length L_D , while T is not zero.

The time-dependent contribution to the phase change reflects the dispersion-induced temporal phase shift that varies with the pulse size $T(z)$ and propagation distance. It influences the temporal and spectral characteristics of the pulse, shaping its overall waveform.

Figure 5 visually illustrates the variation of phase with distance and time in pulse propagation. It provides a clear representation of the transition from positive to negative phase values, showcasing the influence of dispersion on the pulse's temporal phase shift.

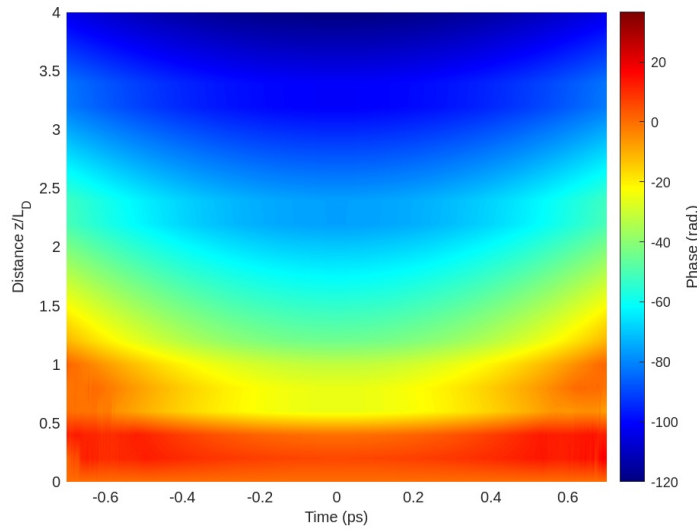


Figure 5: Variation of the Phase with Distance and Time in Pulse Propagation

4.3 Chirp

During pulse propagation, dispersion causes different frequency components of the pulse to travel at different velocities. As a result, the pulse undergoes temporal broadening or narrowing, known as chirp. This chirping effect arises from the dispersion-induced phase shift, which alters the relative phases of the frequency components.

The chirp can be obtained by calculating the gradient of the phase with respect to time. In the provided MATLAB code snippet (Listing 1), the chirp is calculated using the expression:

```
1 phip(1,:) = unwrap(angle(A0));
2 chirp(1,:) = gradient(phip(1,:),TT*1e12)/(2*pi);           % Units of THz
```

Listing 1: The MATLAB code

If we take the gradient of equation (9) with respect to time, the second term vanishes, and we obtain a chirp proportional to $-aT$:

$$\text{Chirp}(z, T) = -\frac{1}{2\pi} \frac{\text{sgn}(\beta_2)(z/L_D)}{1 + (z/L_D)^2} \frac{T}{T(z)^2} \quad (10)$$

Figure 6 illustrates the variation of the chirp with respect to time and distance. It should be noted that some artifacts appear at the boundaries, which we suspect may be related to the MATLAB code.

Upon closer examination of the chirp behavior, two main observations can be made. Firstly, at the center of the pulse (Time = 0), the chirp is always 0. This agrees with equation (10), as for $T = 0$, we have Chirp = 0. Secondly, for a specific distance (e.g., distance = 4), the chirp decreases linearly from 10 to -10, following the $-aT$ relationship.

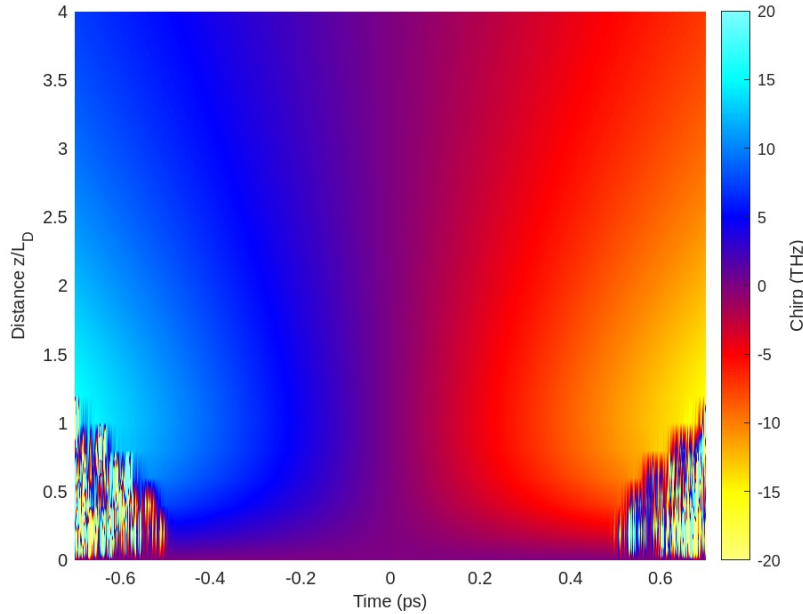


Figure 6: Variation of the Chirp with Distance and Time in Pulse Propagation

4.4 Dispersion Only Output

In the previous sections, we examined the impact of dispersion on various characteristics of the pulse, such as peak power, duration, relative phase, and chirp. Our power analysis revealed that the pulse undergoes stretching, resulting in a decrease in maximum amplitude and an increase in pulse duration. The phase analysis demonstrated a transition from positive to negative phase values as the pulse propagates, indicating the accumulation of dispersion-induced phase shifts. Additionally, the chirp analysis highlighted the presence of temporal chirp, where different parts of the pulse experience varying time delays due to dispersion.

Figure 7 showcases the output pulse after dispersive propagation, illustrating the broadening of the pulse and the decrease in its amplitude. The figure also demonstrates the presence of linear chirp and the quadratic relationship between phase and the temporal size of the pulse.

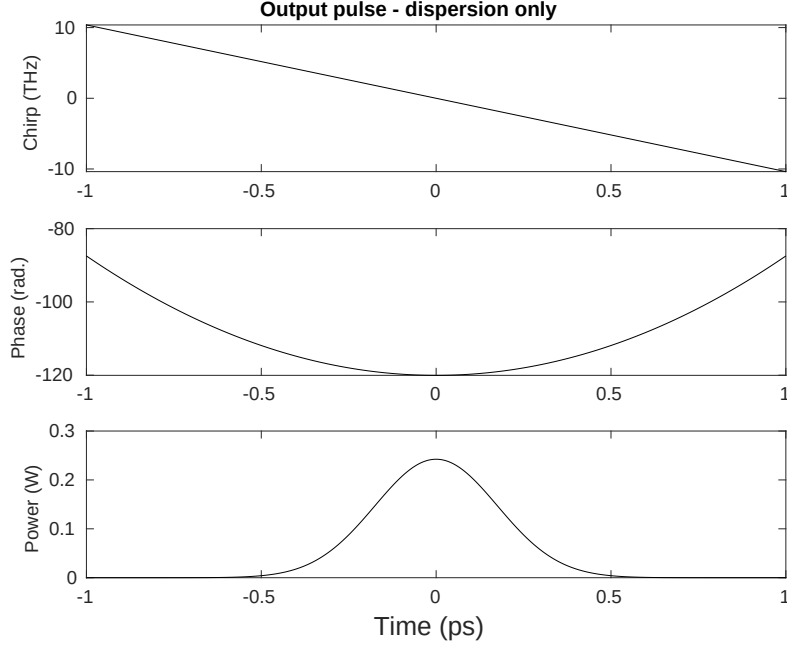


Figure 7: Variation of Chirp, Phase, and Power with Time in the Output Pulse

Here's the revised subsection based on the provided information:

4.5 Dispersion and Nonlinearity

In this section, we examine the simulation results that account for both dispersion and nonlinearity effects in pulse propagation through a nonlinear fiber. When high-intensity optical pulses traverse a nonlinear medium, they undergo a phenomenon known as optical wave breaking or soliton fission, which involves significant spectral broadening and the generation of a wide range of frequencies.

Wave breaking arises due to the interplay between dispersion and nonlinearity. Dispersion causes different frequency components of the pulse to propagate at different speeds, leading to temporal broadening or compression. Nonlinearity, on the other hand, results from the intensity-dependent refractive index of the medium, inducing self-phase modulation and nonlinear frequency conversion.

4.5.1 Power

Figure 8 depicts the power distribution of the pulse in the frequency and temporal domains. As the pulse propagates through the fiber, it undergoes dispersion-induced temporal broadening and frequency separation.

In the temporal domain, the pulse broadens due to dispersion, resulting in a rectangular pulse shape. Simultaneously, in the frequency domain, different frequency components of the pulse spread out, causing frequency separation.

Hence, the power analysis reveals the combined effects of dispersion and nonlinearity, leading to temporal broadening, frequency separation, and the transformation of the pulse shape from Gaussian to rectangular.

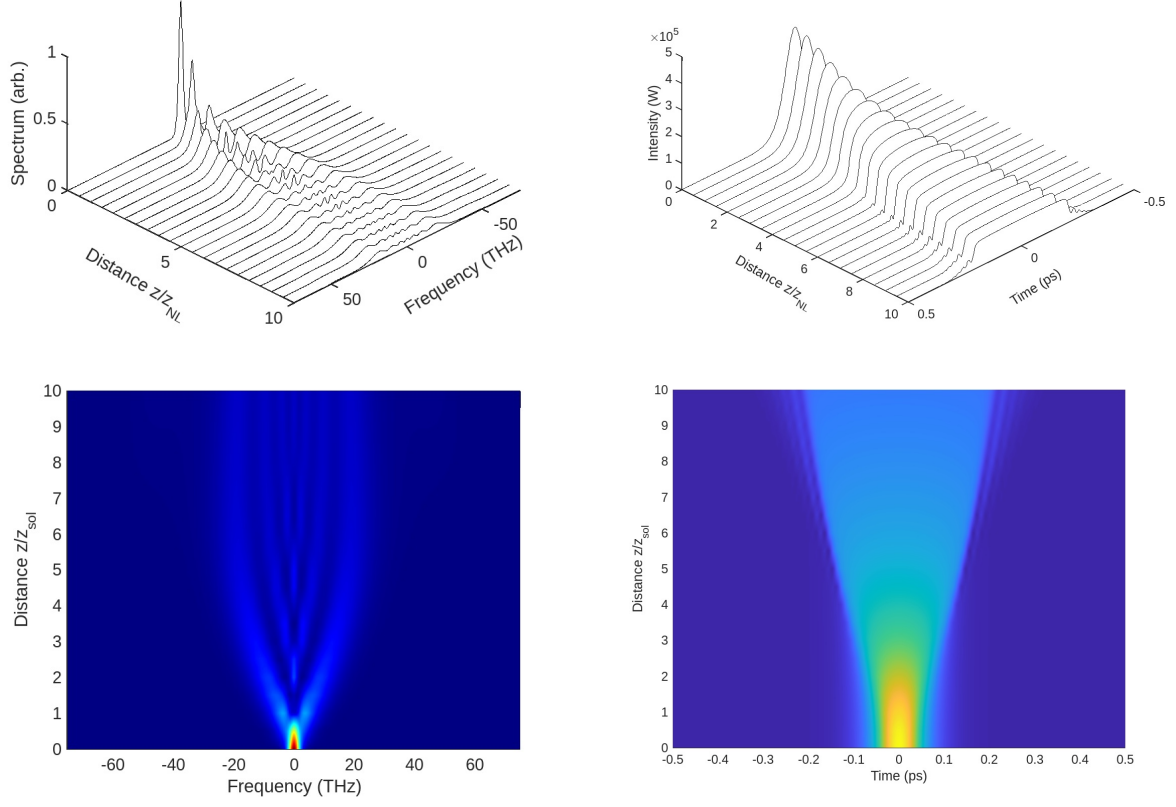


Figure 8: Power distribution in the frequency and temporal domains

4.5.2 Chirp

The evolution of the chirp, representing the variation of phase with respect to time and distance, is presented in Figure 9.

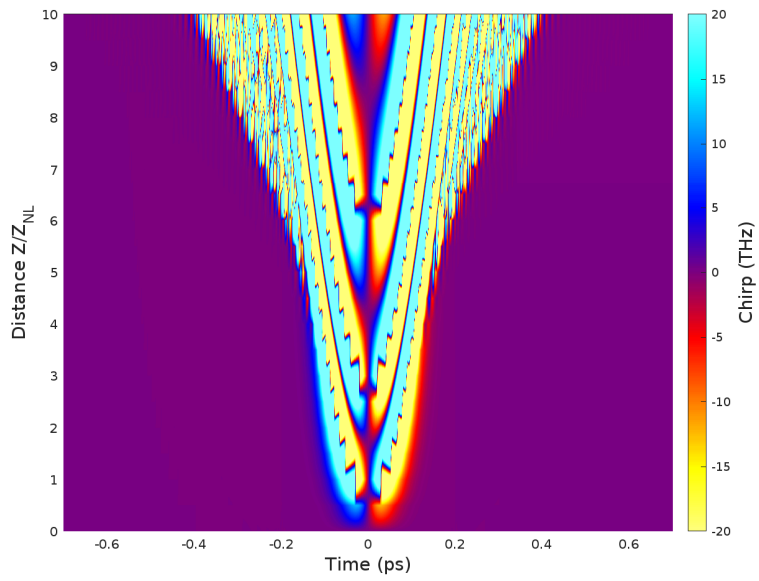


Figure 9: Chirp evolution of the pulse

Figure 9 exhibits a distinctive shape for the chirp variation. Around z/z_{NL} close to zero, the chirp shows a linear behavior, indicating minimal nonlinear effects. Specifically, near time zero, the chirp exhibits a linear variation.

The variation of chirp, phase, and power with time is illustrated in Figure 10, representing the output pulse considering both dispersion and nonlinear effects.

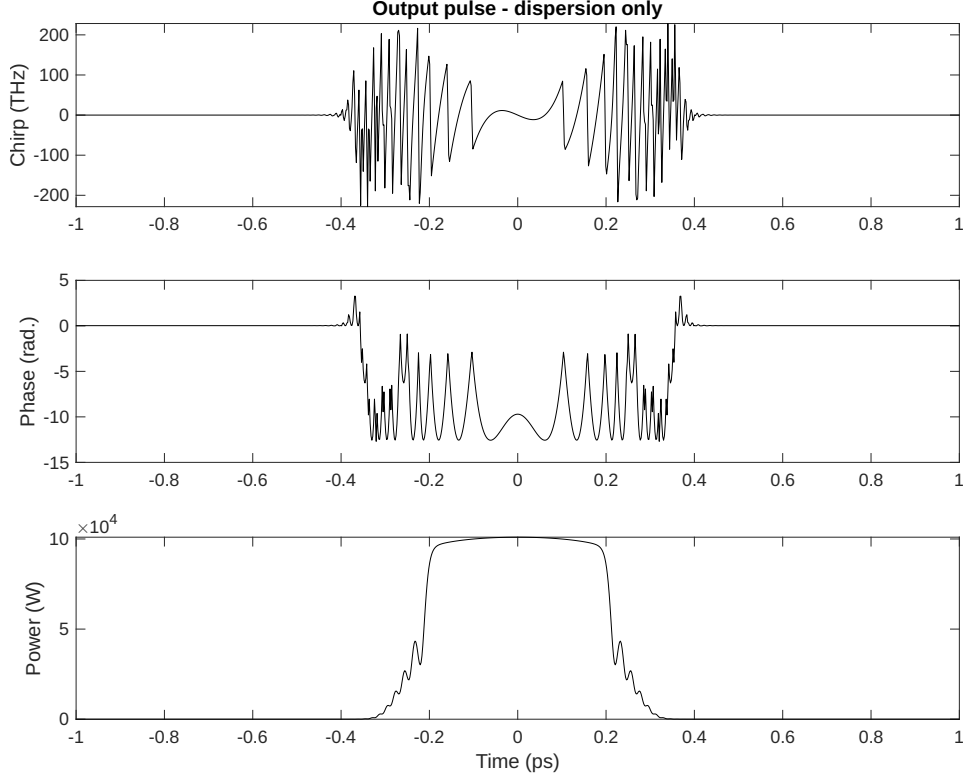


Figure 10: Chirp, phase, and power variation with time

Figure 10 provides further insights into the pulse dynamics by showcasing the variation of chirp, phase, and power with time. The presence of both dispersion and nonlinearity induces complex behaviors in the pulse, resulting in intricate changes in chirp over time.

5 Conclusion

In conclusion, our study examined the impact of dispersion and nonlinearity on the propagation of an optical pulse through a nonlinear fiber. The simulation results revealed important findings.

First, the power analysis showed that dispersion caused the pulse to broaden in the temporal domain, resulting in the separation of frequency components. This broadening transformed the initial rectangular pulse shape into a wider profile.

Second, the chirp analysis demonstrated the interplay between dispersion and nonlinearity. Near the center of the pulse, the chirp showed a linear variation, indicating minimal nonlinear effects. However, as the pulse propagated, the combined influence of dispersion and nonlinearity led to complex changes in the chirp, reflecting the dynamic nature of the pulse's frequency or phase characteristics.

Our study highlights the significance of considering both dispersion and nonlinearity when analyzing pulse propagation. These effects have important implications for applications in optical communications and nonlinear optics.

References

- [1] E. Brookner. “Scientific American 252”. In: 252.2 (1985).
- [2] D. Strickland and G. Mourou. “Compression of amplified chirped optical pulses”. In: (1985).
- [3] Wikipedia. *Scheme of chirped pulse amplification*. 1995. URL: https://commons.wikimedia.org/wiki/File:Chirped_pulse_amplification.png#.